GEORGE WU
(646) 427-2983 ● georgeeewu@gmail.com

SUMMARY

Data Scientist with hands-on experience engineering production-grade machine learning models, end-to-end data pipelines, and generative NLP applications. Proven track record of architecting leak-free validation structures and modular code repositories to transition complex models from research to deployment.

EDUCATION

New York University — New York, NY
B.A. Data Science, minor in mathematics — Sep 2021 - May 2025

- GPA: 3.615/4.
- Applicable coursework and topics: Data Structures(trees, hash table, graphs, Algorithmic complexity (Big O)), Data Management and Analysis(SQL/NoSQL database design, ETL pipelines, data cleaning, and large-scale data manipulation), Fundamentals of Machine Learning(Supervised/Unsupervised Learning, neural networks, decision trees, and model evaluation), Responsible Data Science(Algorithmic fairness, data privacy, AI ethics, and interpretability), Linear Algebra & Probability Theory(Matrix decompositions like SVD/Eigenvalues, vector spaces, stochastic processes, and Bayesian inference), Advanced Topics in Data Science(Deep learning, natural language processing, or time-series forecasting)

University of California, Los Angles — Los Angles, CA
Master in Applied Statistics and Data Science — Sep 2025 - now

- GPA: 4.0/4
- Applicable coursework and topics:Mathematical & Modern Statistics(non-parametric methods, M-estimation, asymptotic theory), Advanced Statistical Communication(Translating complex model outputs into actionable insights, technical writing, and data science interpretability), Deep Learning(backpropagation, optimization, RNNs/Transformers, CNN), Applied Regression(Linear and generalized linear models, variable selection, multicollinearity)

EXPERIENCE

A Round Entertainment — Remote
Machine Learning / Data Science Project manager — Aug 2025 - Apr 2026

- Engineer a two-stage routed regression model using *CatBoost* to predict concert revenue, achieving robust point-estimates by decomposing target variance into separate pricing (Median Log) and attendance share (RMSE) pipelines.
- Designed a dynamic segment-routing architecture that optimized predictions across varying event scales by establishing a 10,000-seat capacity threshold, isolating smaller venues from major stadiums to eliminate model skew.
- Developed a Group-Time aware validation framework to test models grouped by artist to prevent data leakage and evaluate model performance strictly on forward timelines, accurately simulating real-world deployment conditions.
- **Established a standardized MLOps workflow for future engineering teams** by architecting a modular repository footprint, generating dynamic data schemas (columns.json), and centralizing pipeline parameters via YAML configurations.

Sotera Digital Health Remote

Data mining & cleaning Intern Jul 2024 - May 2025

- Engineered optimized real-time patient vital sign streaming data pipelines fetching directly from AWS cloud databases, improving downstream model interpretability and dashboard ingestion rates.
- Collaborated with cross-functional medical and technical teams from Mayo Clinic and SmartAI to format, structure, and clean raw telemetry signals for automated production model integration.
- Organized key features and data formats to better understand patient's vital signs and conditions for downstream analysis.

Industrial Technology Research Institute Hsinchu, Taiwan
Research Intern Jun 2022 - Aug 2022, Jun 2023 - Aug 2023

- Engineered a data pipeline using python tools (e.g. Requests, Beautiful Soup, Pandas) to scrape, extract, and process real-world satellite data.
- Developed a computational framework to analyze geospatial data, successfully reconstructing and inferring consistent satellite trajectory routes.
- Gained production-level exposure to advanced AI paradigms, analyzing the deployment and scaling of reinforcement learning models in industrial environments.
- Programmed and calibrated robotic arms to execute automated and precision tasks, gaining hands-on experience with industrial AI and robotics toolkits.

PROJECTS

CounterFactual storyboard generation New York, NY
Jan 2025 - May 2025

- Fine-tuned Gemma 3-4B to generate counterfactual movie plot summaries using optimized loss functions, cross entropy loss, and hyperparameter tuning; output structured as 10-bullet narrative summaries.
- Engineered a pipeline to feed text outputs into a Stable Diffusion model (runwayML-v1-5), fine-tuned with DreamBooth to achieve a stylized, comic-book aesthetic.
- This project generates narrative-consistent and visually coherent counterfactual storyboards for action movies, enabling screenwriters, educators, and fans to visualize alternate plot trajectories, deepening narrative analysis and providing creative inspiration for film development.

PODS Capstone Project New York, NY
Project Lead Sep 2023 - Dec 2023

- Led a team to analyze a dataset of over 52,000 Spotify songs to model music popularity and define genre-specific audio features.
- Evaluated the relationship of different features within a song, developing a data-driven understanding of music composition.
- Employed statistical tools such as PCA for dimensionality reduction, cross-validation for model validation, and the Mann-Whitney U test to predict popularity outcomes and analyze model performance.

SKILLS

- ***Programming***: Python (PyTorch, Scikit-learn, Beautiful Soup, Requests), R, SQL

- ***Technologies***: AWS(EC2, S3), Jupyter, Git, Docker(Containerization) , Stable Diffusion, DreamBooth, Excel, Hugging Face
- ***Languages***: Mandarin Chinese (Native), English (Native), Korean (Limited Professional Proficiency)
- ***Core Competencies***: Machine Learning, Statistical Modeling, NLP, Data Mining & Cleaning, Data Visualization, Collaborative Teamwork, Communication